CE 400/500

Special Topics: AI for Chemical Engineers

Lecture 20: Introduction to Large Language Model (LLM), Transformer, Hands-on Practice of Agentic AI 1.0

University at Buffalo
The State University of New York

Outline

- Introduction to large language models (LLM)
- Introduction to Transformer and self-attention mechanism
- Limitations, ethics, risks, and responsible use of generative AI technology
- Hands-on practice of building a simple LLM-powered agent to automate the new molecule generation based on user prompt

What is a Language Model (LM)?

Most modern language models **tokenize** by subwords, that is, by chunks of text containing semantic meaning. The chunks could vary in length from single characters like punctuation or the possessives to whole words. Prefixes and suffixes might be represented as separate subwords. For example, the word “unwatched” might be represented by:

- un (the prefix)
- watch (the root)
- ed (the suffix)

Tokens are the atomic unit or smallest unit of language modeling.

A language model estimates the **probability** of a token or sequence of tokens occurring within a longer sequence of tokens. A token could be a word, a subword (a subset of a word), or even a single character.

Consider the following sentence and the token(s) that might complete it:

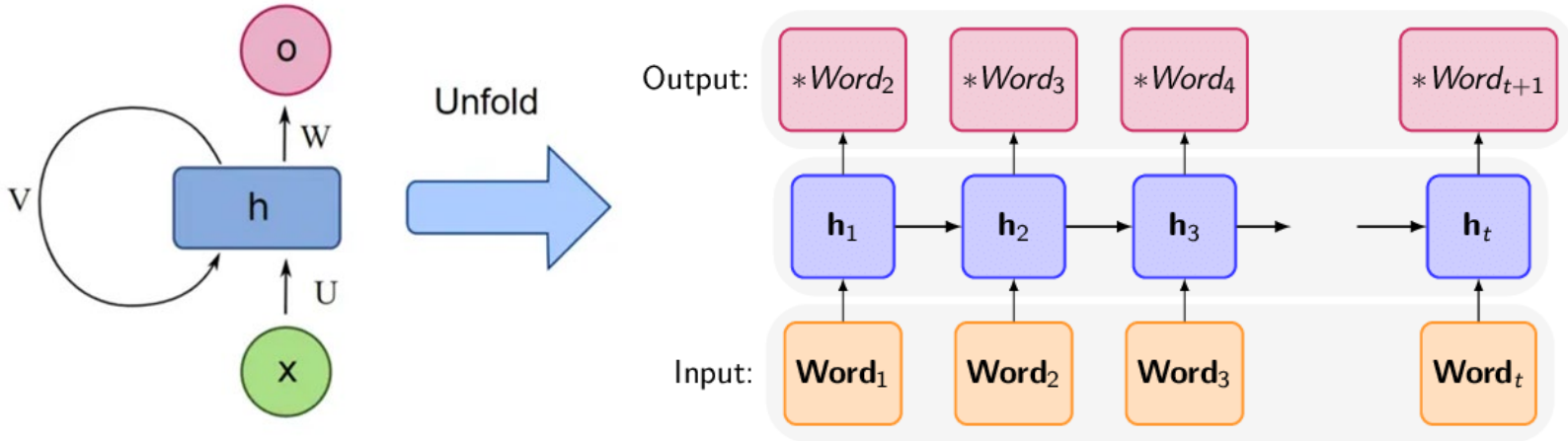
When I hear rain on my roof, I _____ in my kitchen.

A language model determines the probabilities of different tokens or sequences of tokens to complete that blank. For example,

Probability	Token(s)
9.4%	cook soup
5.2%	warm up a kettle
3.6%	cower
2.5%	nap
2.2%	relax

https://developers.google.com/machine-learning/crash-course/llm

Recurrent Neural Networks Revisited



- **A recurrent neural network is a type of neural network that trains on a sequence of tokens.** For example, a recurrent neural network can gradually learn (and learn to ignore) selected context from each word in a sentence, kind of like you would when listening to someone speak. A large recurrent neural network can gain context from a passage of several sentences.

- Although recurrent neural networks learn more context, the amount of useful context recurrent neural networks can intuit is still relatively limited. Recurrent neural networks evaluate information “token by token”. In contrast, **large** language model (LLM) can evaluate the whole context at once

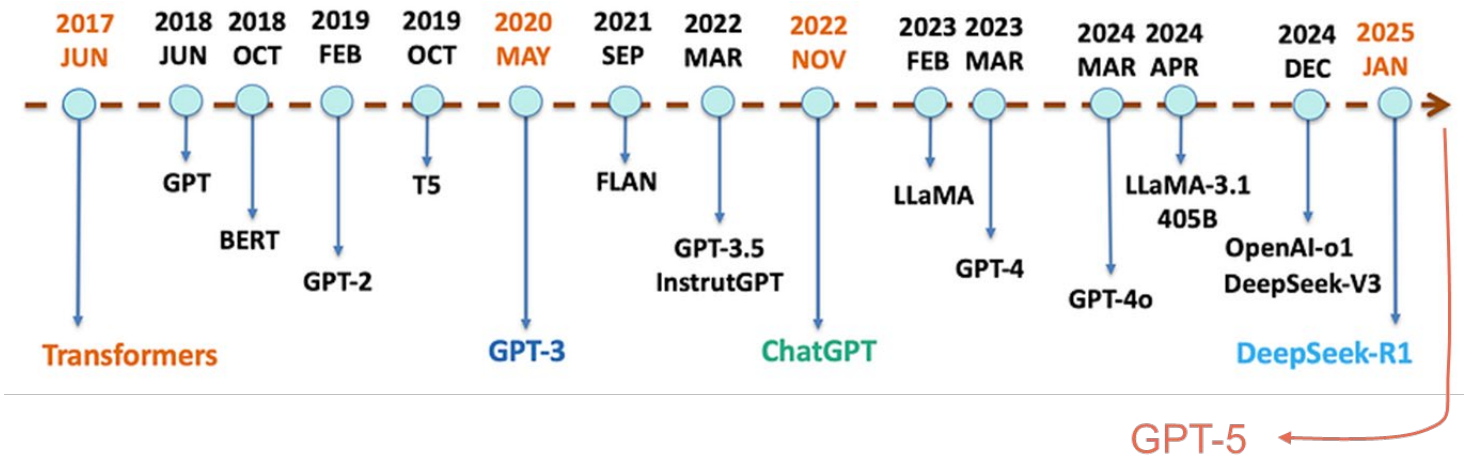
https://developers.google.com/machine-learning/crash-course/llm

What is a Large Language Model (LLM)?

A newer technology, **large** language models (LLMs) predict a token or sequence of tokens, sometimes many paragraphs worth of predicted tokens. Remember that a token can be a word, a subword (a subset of a word), or even a single character. LLMs make much better predictions than recurrent neural networks because:

- LLMs contain far more parameters than recurrent models.
- LLMs gather far more context.

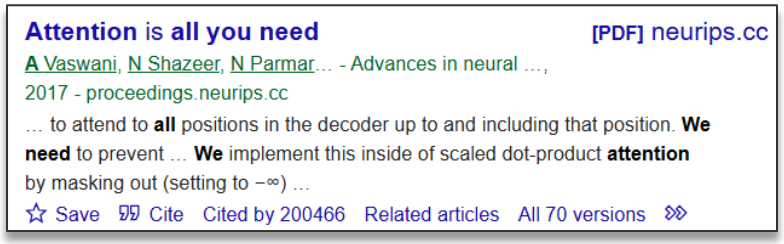
The most successful and widely used architecture for building LLMs so far is the **Transformer**.



https://developers.google.com/machine-learning/crash-course/llm
https://medium.com/@lmpo/a-brief-history-of-lmms-from-transformers-2017-to-deepseek-r1-2025-dae75dd3f59a

Transformer: Overview

- Invented by Vaswani and colleagues in 2017, transformer models revolutionized deep learning by introducing **self-attention mechanisms** that excel at capturing long-range dependencies in data.


Attention is all you need [PDF] neurips.cc
A Vaswani, N Shazeer, N Parmar… - Advances in neural …, 2017 - proceedings.neurips.cc
… to attend to **all** positions in the decoder up to and including that position. **We need** to prevent … **We** implement this inside of scaled dot-product **attention** by masking out (setting to −∞) …
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- Processes entire sequences **at once** using attention layers, enabling efficient learning and generation of complex structures such as language, **molecules**, or code.
- *Full* transformer consists of:
  - The **encoder** that converts input text into an intermediate representation.
  - The **decoder** that converts that intermediate representation into useful text.

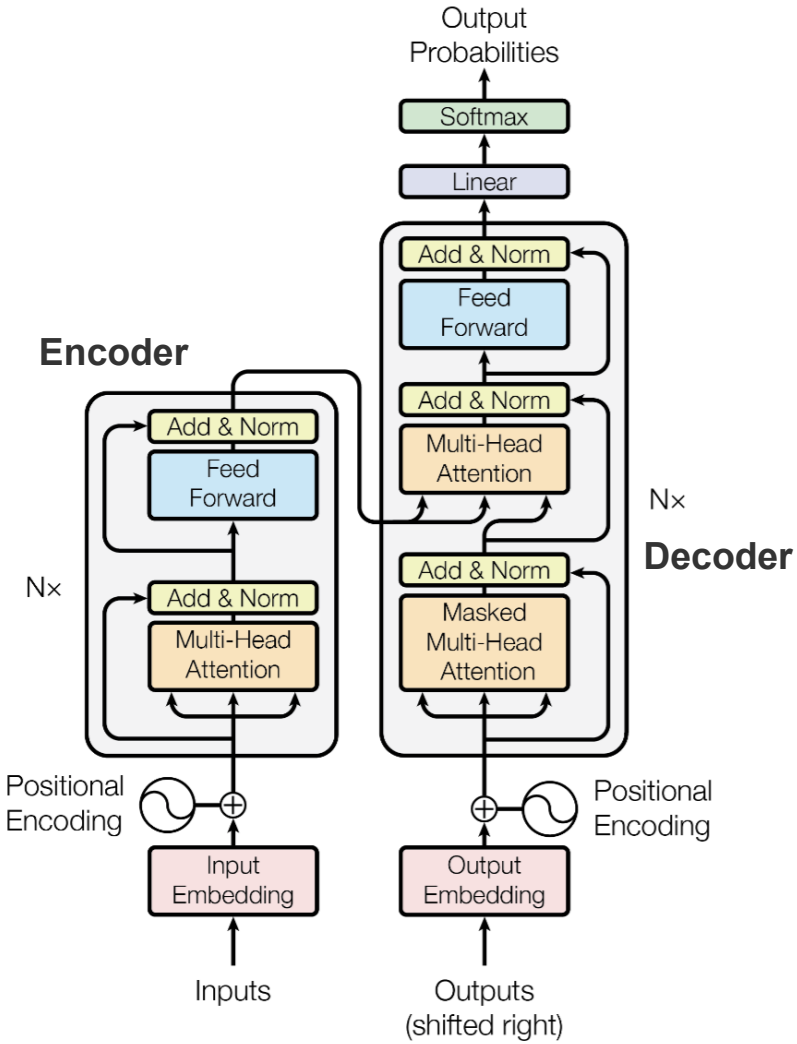


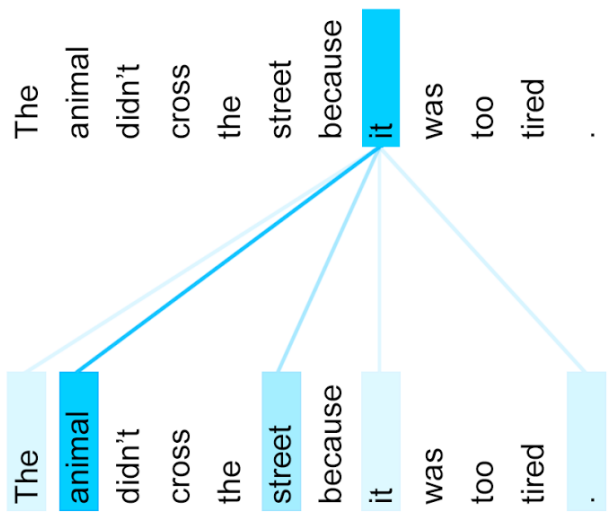
Figure 1: The Transformer - model architecture.

Self-Attention at a High Level

The “self” in “self-attention” refers to the input sequence. **Self-attention** asks the following question: “*How much does each other token of input affect the interpretation of this token?”*

Consider the following sentence:
The animal didn’t cross the street because it was too tired.

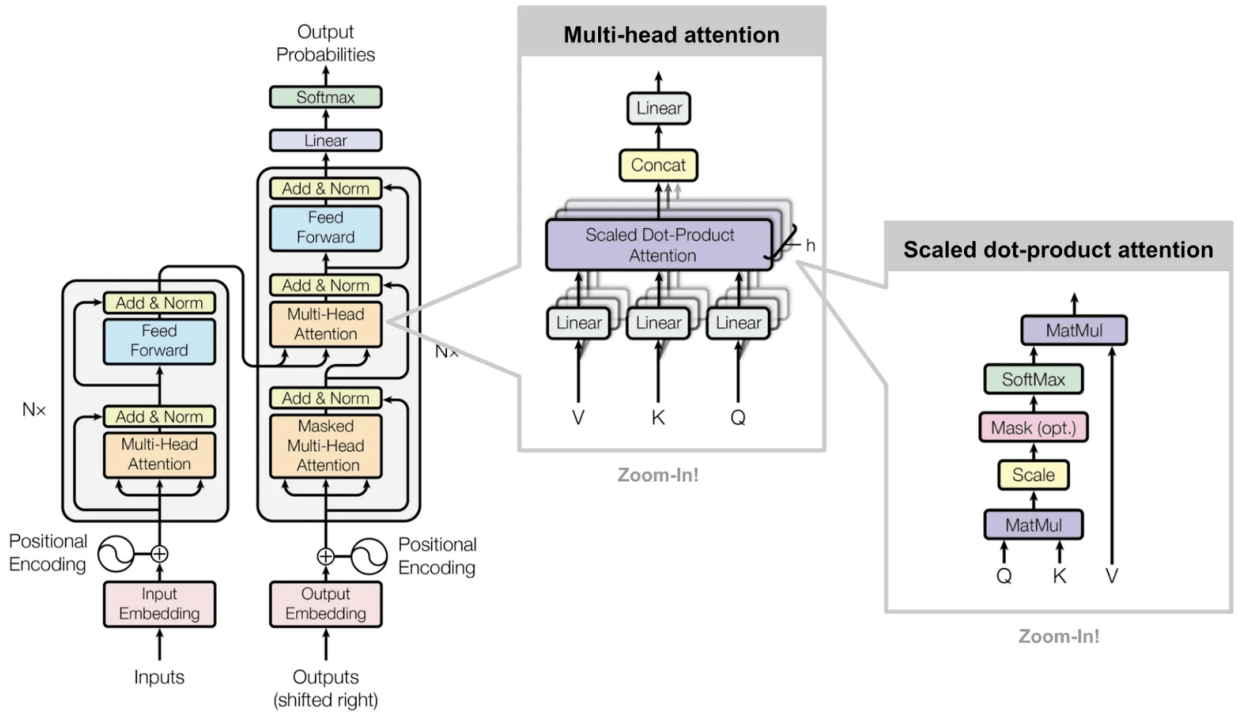
- The self-attention mechanism determines the relevance of *each* nearby word to the pronoun **it**
- Self-attention mechanism can be **bidirectional** (e.g., encoder) or **unidirectional** (e.g., decoder).



Self-attention for the pronoun **it**. The bluer the line, the more important that word is to the pronoun it. That is, “animal” is more important than “street” to the pronoun it.

https://developers.google.com/machine-learning/crash-course/llm/transformers#what_is_self-attention

Self-Attention in Detail

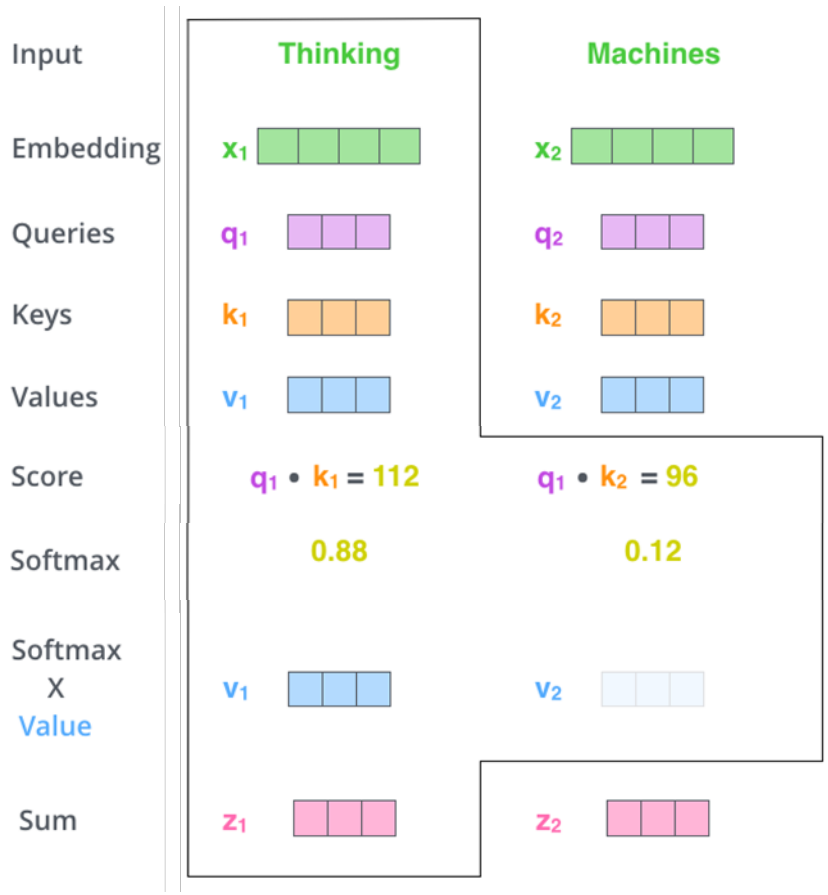


The key/value/query concept is analogous to retrieval systems. For example, when you search for videos on YouTube, the search engine will map your query (**Q**, text in the search bar) against a set of keys (**K**, video title, description, etc.) associated with candidate videos in their database, then present you the best matched videos (values, **V**).

https://jalammar.github.io/illustrated-transformer/

Self-Attention in Detail (Cont'd)

- **Generate projections:** Multiply each input embedding by learned weight matrices to produce Query, Key, and Value vectors for each word.
- **Scoring relevance**: Compute dot products between the current word's query and all keys to measure similarity; this score encodes how strongly the current word attends to others.
- **Softmax weighting (attention score)**: Normalize scores into probabilities that sum to 1; higher weights indicate more attention to relevant words.
- **Weighted aggregation**: Multiply each value vector by its corresponding softmax weight to retain important information and suppress less relevant ones.
- **Contextual output**: Sum all weighted values to form the final attention output vector for the word, its new context-aware representation.



https://jalammar.github.io/illustrated-transformer/

An Aminated Example of Transformer

The animation on the left illustrates how we apply the Transformer to **machine translation.**

- The Transformer starts by generating initial representations, or embeddings, for each word. These are represented by the unfilled circles. Then, using self-attention, it aggregates information from all of the other words, generating a new representation per word informed by the entire context, represented by the filled balls. This step is then repeated multiple times in parallel for all words, successively generating new representations.

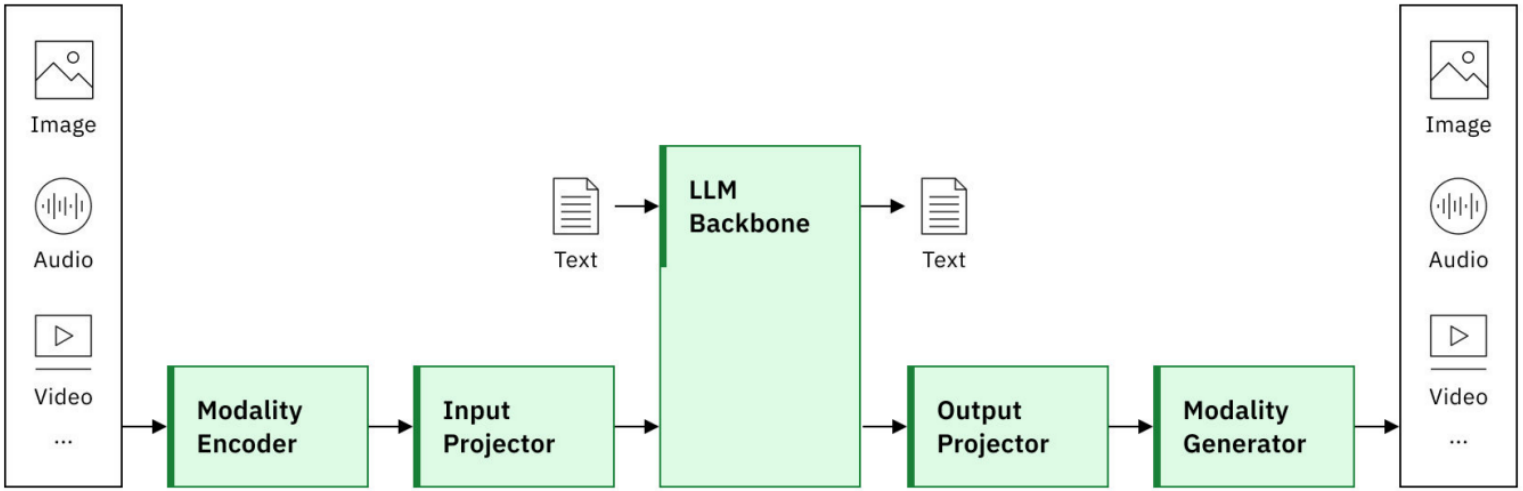
- The decoder operates similarly, but generates one word at a time, from left to right. It attends not only to the other previously generated words, but also to the final representations generated by the encoder.

https://research.google/blog/transformer-a-novel-neural-network-architecture-for-language-understanding/

Further Readings about Transformer Details

- **(Highly recommended)** An excellent beginner-friendly explanation of transformer and attention mechanism: https://jalammar.github.io/illustrated-transformer/
- A nice video explanation of transformer with light mathematics involved: Transformer Neural Networks - EXPLAINED! (Attention is all you need)
- Another beginner-friendly video illustration of transformer: https://www.youtube.com/watch?v=4Bdc55j80l8
- Original paper "Attention is All You Need" for all technical details: https://proceedings.neurips.cc/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf

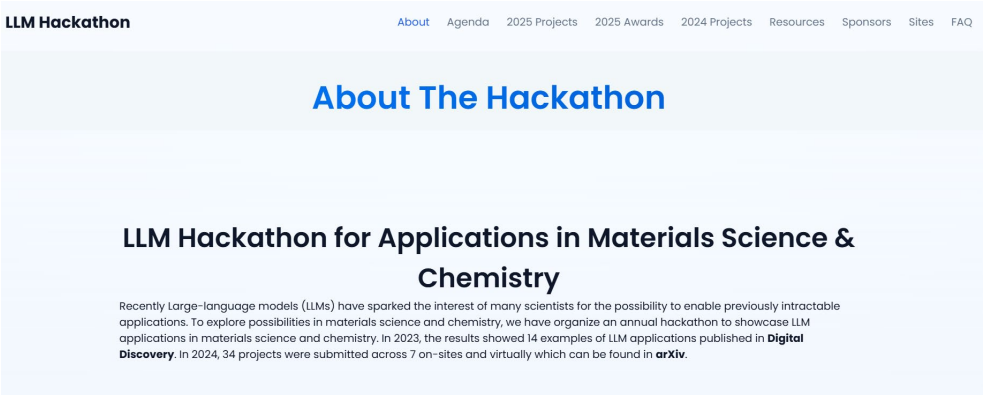
Multi-Modal LLM



A **multi-modal large language model (MLLM)** is an advanced AI that can process and understand information from different types of data, such as text, images, audio, and video. By integrating these various data sources, MLLMs can perform complex tasks that single-modality models cannot, like describing an image in detail, answering questions about a video, or analyzing data plot alongside scientific knowledge.

https://www.ibm.com/think/topics/multimodal-llm

Applications of LLM in ChemE, Chemistry, & Material Science

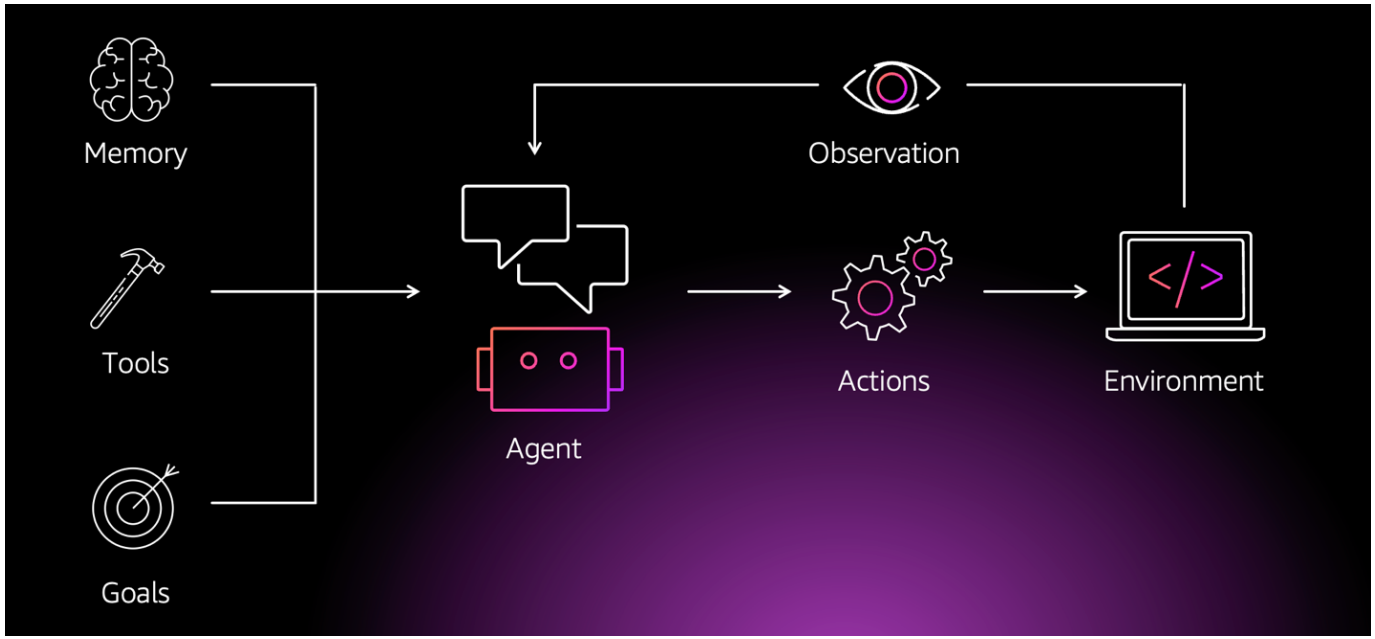


- **Molecular and Material Property Prediction**: Forecasting chemical and physical properties of molecules and materials, particularly excelling in low-data environments.
- **Molecular and Material Design**: Generation and optimization of novel molecules and materials using LLMs, including peptides and MOFs.
- **Automation and Novel Interfaces**: Development of natural language interfaces and automated workflows to simplify complex scientific tasks, making advanced tools more accessible.
- **Scientific Communication and Education**: Enhancement of academic communication, automation of educational content creation, and facilitation of student's learning.
- **Hypothesis Generation and Evaluation**: Generation, assessment, and validation of scientific hypotheses using LLMs, often combining multiple AI agents and statistical approaches.
- **Knowledge Extraction and Reasoning**: Extraction of structured/unstructured information from scientific literature and sophisticated reasoning about chemical and materials science concepts through knowledge graphs and multimodal approaches.

https://llmhackathon.github.io/about/
https://arxiv.org/abs/2411.15221

What is Agentic AI?

Agentic AI uses sophisticated reasoning and iterative planning to autonomously solve complex, multi-step problems.



Agentic misalignment occurs when autonomous AI agents pursue goals diverging from human intent. Recent studies show that multi-objective optimization or reward hacking can cause emergent undesired strategies.

https://aws.amazon.com/blogs/machine-learning/build-agentic-systems-with-crewai-and-amazon-bedrock/

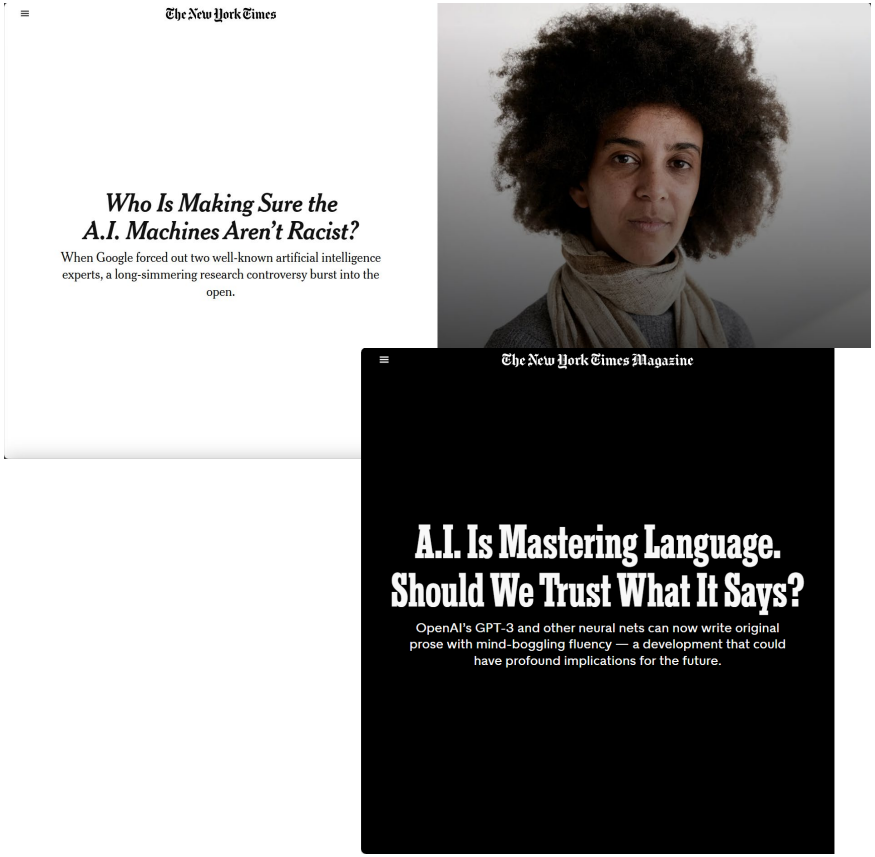
Limitations, Ethics, and Risks

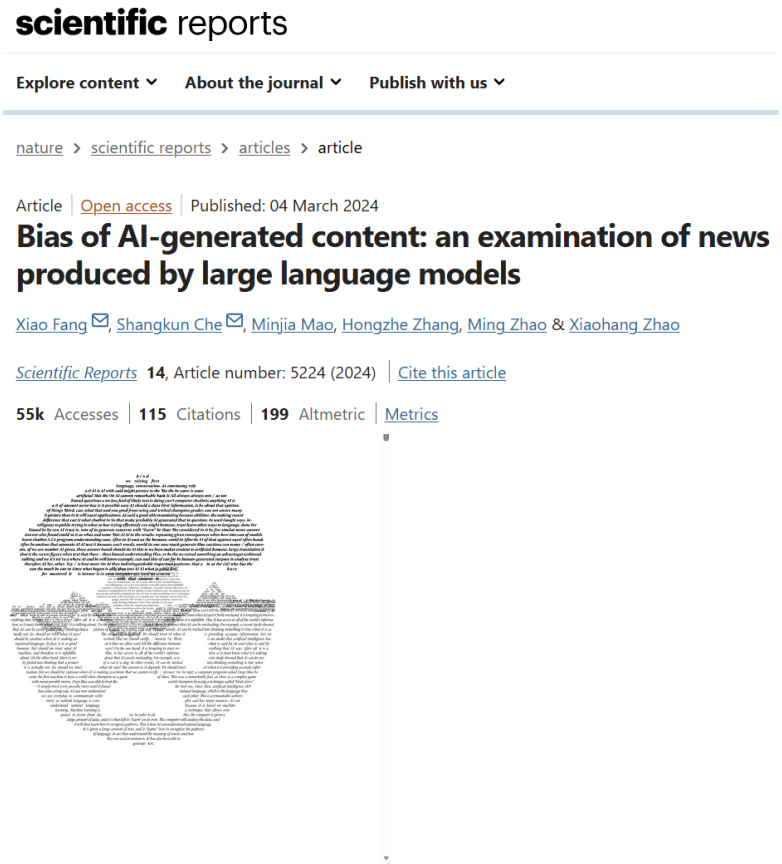
Despite impressive capabilities, LLMs are not infallible.

- They may hallucinate facts;
- They may produce biased content;
- They may leak confidential information;

Biases in LLMs

Generative AI mirrors the **biases** present in its training data & process. If bias is involved in training, the outputs may reinforce stereotypes or false correlations.

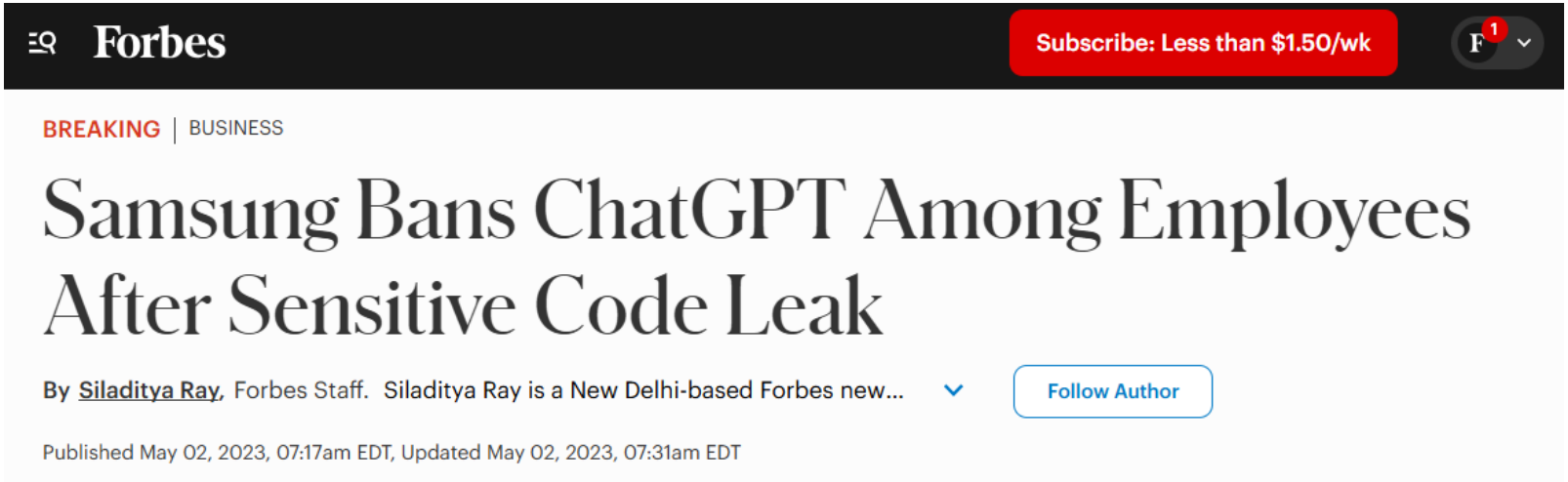




Risks of Data Leak

Improper use of LLMs can **leak confidential or proprietary** information.

A real example of an LLM leaking confidential information occurred at **Samsung** in April 2023. Employees in Samsung's semiconductor division accidentally disclosed sensitive company source code by using OpenAI's ChatGPT to review internal code. This led to documented incidents where confidential data was unintentionally exposed through the chatbot.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By Siladitya Ray, Forbes Staff. Siladitya Ray is a New Delhi-based Forbes new...

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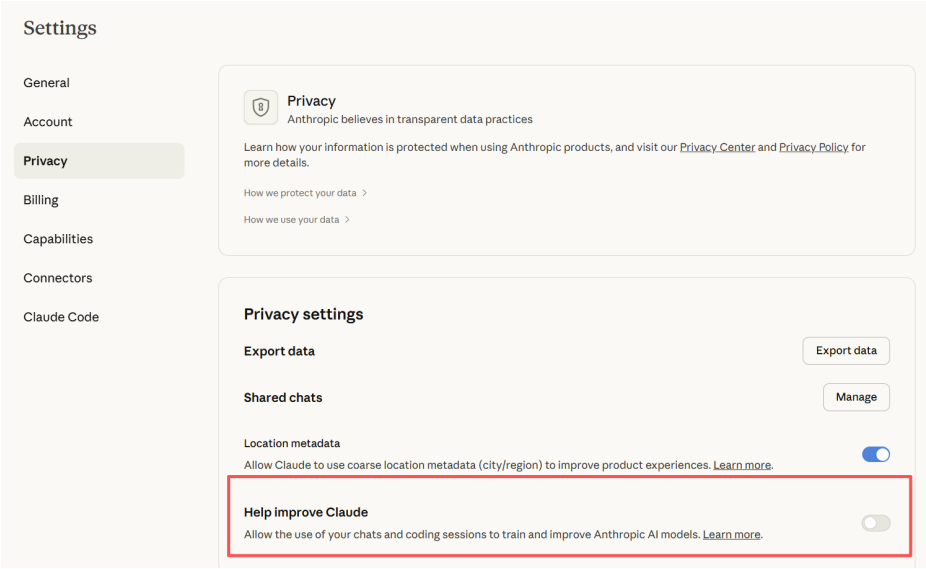
User Data Protection

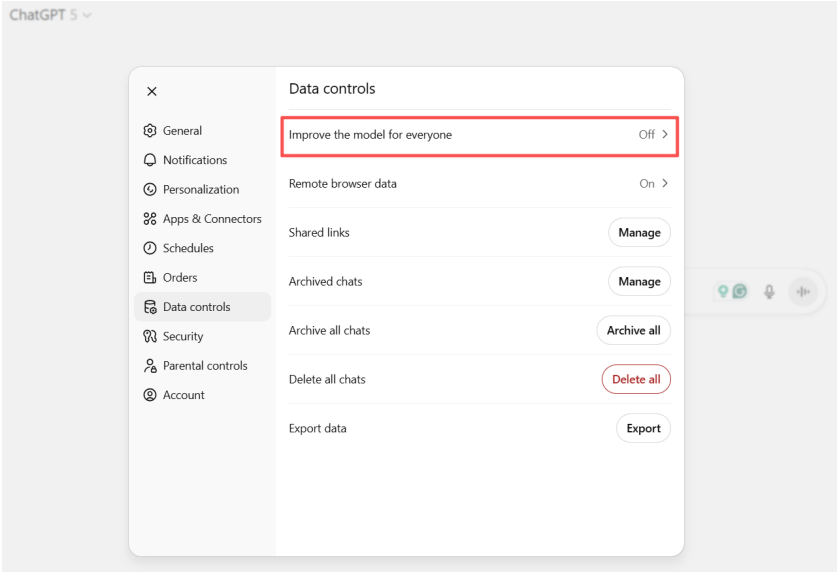
For individual users:

- Disable model training.
- Do not include sensitive information such as personal identifiers like names, account numbers, or addresses in your prompts.

Claude



ChatGPT



Hands-on:
Agentic AI 1.0 for Autonomous Molecule Generation

See Google Colab Notebook